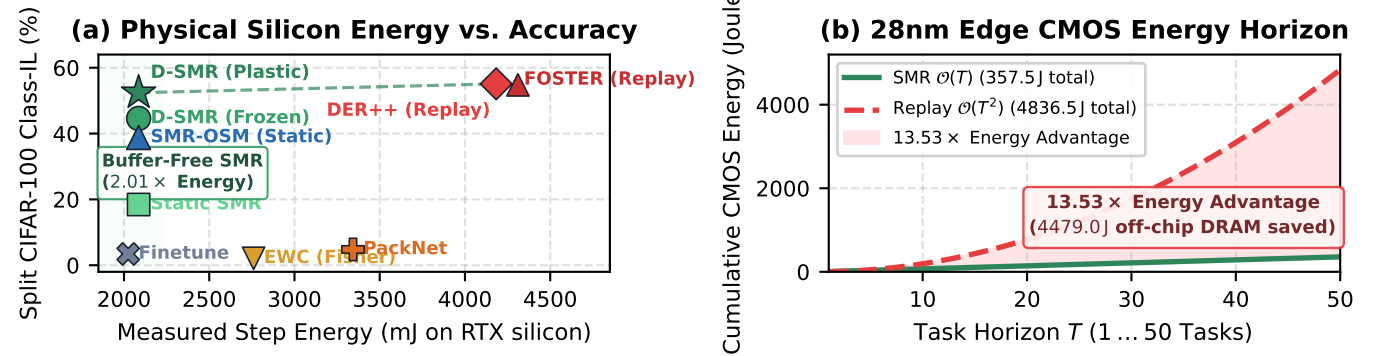
(a) Physical Silicon Energy vs. Accuracy
Split CIFAR-100 Class-IL (%)
Measured Step Energy (mJ on RTX silicon)
D-SMR (Plastic)
FOSTER (Replay)
DER++ (Replay)
D-SMR (Frozen)
SMR-OSM (Static)
Buffer-Free SMR
(2.01× Energy)
Static SMR
Finetune
EWC (Fisher)
PackNet
(b) 28nm Edge CMOS Energy Horizon
Cumulative CMOS Energy (Joule
SMR $\mathcal{O}(T)$ (357.5 J total)
Replay $\mathcal{O}(T^2)$ (4836.5 J total)
13.53× Energy Advantage
13.53× Energy Advantage
(4479.0 J off-chip DRAM saved)
Task Horizon T (1 … 50 Tasks)